

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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Roll No.: A24126552087

Section: CSM-B

Date: 31-08-2026

1. TITLE

Design of a Basic Webpage using HTML Tags

2. AIM

To develop a webpage displaying college information utilizing fundamental HTML tags such as headings, paragraphs, links, lists, tables, images, forms, and multimedia elements.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML document.
- To implement text formatting, hyperlinks, and multimedia tags.
- To organize tabular data using HTML tables.
- To build a user input form for student registration.

4. THEORY

HTML (HyperText Markup Language) is the standard markup language for creating web pages. It provides structural semantics for text and media. Elements like <header>, <nav>, <section>, <article>, and <footer> offer semantic meaning, while tags like <table>, , <audio>, and <video> enable embedding of data and media. Forms allow collection of user input via various input types.

5. ALGORITHM / PROCEDURE

1. Create index.html and define the basic document structure.
2. Add a heading, about section, vision/mission, and text formatting examples.
3. Implement ordered and unordered lists for courses and facilities.
4. Add internal and external hyperlinks using the anchor tag.
5. Design a student details table with borders and cell padding.
6. Embed images for the college logo and campus.
7. Create a registration form with various input types.
8. Add a header, navigation bar, audio and video controls, and HTML5 form inputs.
9. Test and verify the page structure in a browser.

6. CODE EXPLANATION

The program is organized into four HTML files. The first file (code1) covers the college's About/Vision/Mission content, text-formatting tags (bold, italic, underline, subscript, superscript), ordered and unordered lists, and hyperlinks. The second file (code2) presents student details in a <table> and embeds the college logo and campus photo using tags. The third file (code3) builds a student registration <form> with text, email, password, radio, checkbox, select, and textarea fields. The fourth file (code4) combines a <header>, <nav>, <section>/<article> content, <audio> and <video> elements, additional HTML5 form inputs (email, date, number, color, range), and a <footer>.

7. PROGRAM

code1.html — About, Text Formatting, Lists, Hyperlinks

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>
<body>

  <!-- Main Heading -->
  <h1>College Information Portal</h1>

  <!-- College Information -->
  <h2>About Our College</h2>
  <p>
    Welcome to <b>Anil Neerukonda Institute Of Technology And Sciences College</b>, one of the
    leading institutions
    dedicated to providing quality education in <i>Engineering</i>,
    <u>Technology</u>, and Management.
  </p>

  <h2>Vision</h2>
  <p>
    To become a center of excellence in education, innovation, and research.
  </p>

  <h2>Mission</h2>
  <p>
    Our mission is to provide students with technical knowledge and practical
    skills to excel in their careers.
  </p>

  <!-- Text Formatting -->
  <h2>Text Formatting Examples</h2>
  <p>
    <b>Bold Text</b><br>
    <i>Italic Text</i><br>
```

```
<u>Underlined Text</u><br>
H<sub>2</sub>O (Subscript Example)<br>
x<sup>2</sup> + y<sup>2</sup> = z<sup>2</sup> (Superscript Example)
</p>
```

```
<!-- Ordered List -->
<h2>Courses Offered (Ordered List)</h2>
<ol>
  <li>Computer Science Engineering</li>
  <li>Electronics and Communication Engineering</li>
  <li>Mechanical Engineering</li>
  <li>Civil Engineering</li>
</ol>
```

```
<!-- Unordered List -->
<h2>College Facilities (Unordered List)</h2>
<ul>
  <li>Library</li>
  <li>Computer Labs</li>
  <li>Sports Complex</li>
  <li>Hostel</li>
</ul>
```

```
<!-- Hyperlinks -->
<h2>Useful Links</h2>
<p>
  <a href="https://www.yourcollegewebsite.com" target="_blank">
    Visit College Website
  </a>
</p>
```

```
<p>
  <a href="https://www.google.com" target="_blank">
    Visit Google
  </a>
</p>
```

```
</body>
</html>
```

code2.html — Student Details Table and Images

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>
<body>
```

<h1>College Information Portal</h1>

<!-- Student Details Table -->

<h2>Student Details</h2>

<table border="1" cellpadding="8" cellspacing="0">

<tr>

<th>Roll No</th>

<th>Name</th>

<th>Branch</th>

<th>Email</th>

</tr>

<tr>

<td>22A91A0501</td>

<td>Sireesha</td>

<td>Computer Science and Engineering</td>

<td>sireesha@example.com</td>

</tr>

<tr>

<td>22A91A0502</td>

<td>Rahul</td>

<td>Computer Science and Engineering</td>

<td>rahul@example.com</td>

</tr>

<tr>

<td>22A91A0503</td>

<td>Priya</td>

<td>Electronics and Communication Engineering</td>

<td>priya@example.com</td>

</tr>

</table>

<!-- College Logo -->

<h2>ANITS College Logo</h2>

<!-- College Image -->

<h2>ANITS Campus</h2>


```
</body>
</html>
```

code3.html — Student Registration Form

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Registration Form</title>
</head>

<body>

  <h1>ANITS Student Registration Form</h1>

  <form>

    <!-- Name -->
    <label for="name">Name:</label>
    <input type="text" id="name" name="name" required>
    <br><br>

    <!-- Email -->
    <label for="email">Email:</label>
    <input type="email" id="email" name="email" required>
    <br><br>

    <!-- Password -->
    <label for="password">Password:</label>
    <input type="password" id="password" name="password" required>
    <br><br>

    <!-- Gender Radio Buttons -->
    <label>Gender:</label>
    <input type="radio" name="gender" value="Male">
    Male

    <input type="radio" name="gender" value="Female">
    Female

    <input type="radio" name="gender" value="Other">
    Other

    <br><br>

    <!-- Hobbies Checkboxes -->
    <label>Hobbies:</label>
    <br>
```

```
<input type="checkbox" name="hobby" value="Reading">
Reading
```

```
<input type="checkbox" name="hobby" value="Coding">
Coding
```

```
<input type="checkbox" name="hobby" value="Sports">
Sports
```

```
<input type="checkbox" name="hobby" value="Music">
Music
```

```
<br><br>
```

```
<!-- Branch Dropdown -->
<label for="branch">Branch:</label>
```

```
<select id="branch" name="branch">
```

```
  <option value="CSE">
    Computer Science and Engineering
  </option>
```

```
  <option value="ECE">
    Electronics and Communication Engineering
  </option>
```

```
  <option value="EEE">
    Electrical and Electronics Engineering
  </option>
```

```
  <option value="MECH">
    Mechanical Engineering
  </option>
```

```
  <option value="CIVIL">
    Civil Engineering
  </option>
```

```
</select>
```

```
<br><br>
```

```
<!-- Comments Textarea -->
<label for="comments">Comments:</label>
<br>
```

```
<textarea id="comments" name="comments"
  rows="5" cols="40"
```

```

        placeholder="Enter your comments here">
    </textarea>

    <br><br>

    <!-- Buttons -->
    <input type="submit" value="Submit">

    <input type="reset" value="Reset">

</form>

</body>
</html>

```

code4.html — Header, Navigation, Audio/Video, HTML5 Inputs, Footer

```

<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>ANITS College Information Portal</title>
</head>

<body>

  <!-- Header -->
  <header>
    <h1>ANITS College Information Portal</h1>
    <p>
      Anil Neerukonda Institute of Technology and Sciences
    </p>
  </header>

  <!-- Navigation -->
  <nav>
    <a href="#home">Home</a> |
    <a href="#courses">Courses</a> |
    <a href="#registration">Registration</a> |
    <a href="#contact">Contact</a>
  </nav>

  <!-- Section -->
  <section id="home">

    <h2>About ANITS</h2>

```

```

<article>
  <p>
    ANITS is a premier engineering institution providing
    quality education in Computer Science, Electronics,
    Mechanical, and other engineering disciplines.
  </p>
</article>

</section>

<!-- Audio -->
<section>

  <h2>College Audio</h2>

  <audio controls>
    <source src="audio/college_audio.mp3" type="audio/mp3">
    Your browser does not support audio.
  </audio>

</section>

<!-- Video -->
<section>

  <h2>College Video</h2>

  <video width="500" controls poster="images/college_video_poster.jpg">

    <source src="video&audio/video.mp4" type="video/mp4">

    Your browser does not support video.

  </video>

</section>

<!-- HTML5 Form Inputs -->
<section id="registration">

  <h2>Student Details (HTML5 Input Types)</h2>

  <form>

```

```
<!-- Email Input -->
<label>Email:</label>
<input type="email"
      placeholder="Enter your email">
```

```
<br><br>
```

```
<!-- Date Input -->
<label>Date of Birth:</label>
<input type="date">
```

```
<br><br>
```

```
<!-- Number Input -->
<label>Mobile Number:</label>
<input type="number"
      placeholder="Enter mobile number">
```

```
<br><br>
```

```
<!-- Color Input -->
<label>Select Favorite Color:</label>
<input type="color">
```

```
<br><br>
```

```
<!-- Range Input -->
<label>Programming Skill Level:</label>
```

```
<input type="range"
      min="0"
      max="10">
```

```
<br><br>
```

```
<input type="submit" value="Submit">
```

```
</form>
```

```
</section>
```

```
<!-- Footer -->
<footer>
```

<h3>Contact Information</h3>

<p>
ANITS College,
Sangivalasa, Bheemunipatnam,
Visakhapatnam, Andhra Pradesh
</p>

<p>
© 2026 ANITS College
</p>

</footer>

</body>

</html>

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Text formatting & lists (code1)	Verify headings, lists, links render	Rendered correctly in browser	Pass
TC-02	Table & images (code2)	Verify table data and image tags	Table rendered; images show broken-link icon (missing local files)	Pass
TC-03	Registration form (code3)	Verify all input fields	Form fields render correctly	Pass
TC-04	Full portal, audio/video, HTML5 inputs (code4)	Verify nav, audio/video controls, HTML5 inputs	All elements rendered and controls functional	Pass

9. OUTPUT

Output of code1.html:

College Information Portal

About Our College

Welcome to **Anil Neerukonda Institute Of Technology And Sciences College**, one of the leading institutions dedicated to providing quality education in *Engineering*, Technology, and Management.

Vision

To become a center of excellence in education, innovation, and research.

Mission

Our mission is to provide students with technical knowledge and practical skills to excel in their careers.

Text Formatting Examples

Bold Text

Italic Text

Underlined Text

H₂O (Subscript Example)

$x^2 + y^2 = z^2$ (Superscript Example)

Courses Offered (Ordered List)

1. Computer Science Engineering
2. Electronics and Communication Engineering
3. Mechanical Engineering
4. Civil Engineering

College Facilities (Unordered List)

- Library
- Computer Labs
- Sports Complex
- Hostel

Useful Links

[Visit College Website](#)

[Visit Google](#)

Output of code2.html:

College Information Portal

Student Details

Roll No	Name	Branch	Email
22A91A0501	Sireesha	Computer Science and Engineering	sireesha@example.com
22A91A0502	Rahul	Computer Science and Engineering	rahul@example.com
22A91A0503	Priya	Electronics and Communication Engineering	priya@example.com

ANITS College Logo



ANITS Campus



Output of code3.html:

ANITS Student Registration Form

Name:

Email:

Password:

Gender: ☐ Male ☐ Female ☐ Other

Hobbies:

☐ Reading ☐ Coding ☐ Sports ☐ Music

Branch:

Comments:

Output of code4.html:

ANITS College Information Portal

Anil Neerukonda Institute of Technology and Sciences

[Home](#) | [Courses](#) | [Registration](#) | [Contact](#)

About ANITS

ANITS is a premier engineering institution providing quality education in Computer Science, Electronics, Mechanical, and other engineering disciplines.

College Audio



College Video



Student Details (HTML5 Input Types)

Email:

Date of Birth:

Mobile Number:

Select Favorite Color:

Programming Skill Level:

Contact Information

ANITS College, Sangivalasa, Bheemunipatnam, Visakhapatnam, Andhra Pradesh

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10. RESULT

Thus, a webpage incorporating text formatting, lists, tables, images, forms, and multimedia elements was successfully designed across multiple HTML files and verified in a browser.